Extreme Subgroup Effects Under a Uniform Treatment Benefit

A Simulation Study Based on Observed Trial Data

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Section 1

Motivation & Overview

The Core Question

Subgroup analysis in regulatory submissions

When many subgroups are examined in a single trial, small subgroups can produce hazard ratio estimates that look dramatic *purely by chance*.

This study asks directly

Under a DGM with a **perfectly uniform treatment benefit** (true HR $=$ 0.70 for every patient), how often do standard Cox model analyses produce extreme subgroup results?

- Clinical subgroups of a given N are statistically indistinguishable from meaningless *random* groupings of the same size
- The **random** benchmarks random combinations and should fluctuate around overall results with "extremeness" due to variability of small samples as opposed to true subgroup effects

Setting: Uniform Treatment Effect

The DGM is built from a real trial (GBSG breast cancer, $N = 686$) using `generate_aft_dgm_flex()`.

Key feature — constant log hazard ratio

$$\psi^0(\mathbf{L}) = \log(\mathrm{HR}_{\mathrm{true}}) \quad \text{for all } \mathbf{L}.$$

- True overall HR $=$ 0.70 (moderate benefit)
- Every subgroup shares *exactly* this HR
- Any apparent subgroup finding is a false positive by construction

Spline parameterisation

`log.hrs = log(c(0.70, 0.70, 0.70))` — uniform HR at biomarker values 0, knot k, and upper anchor ζ.

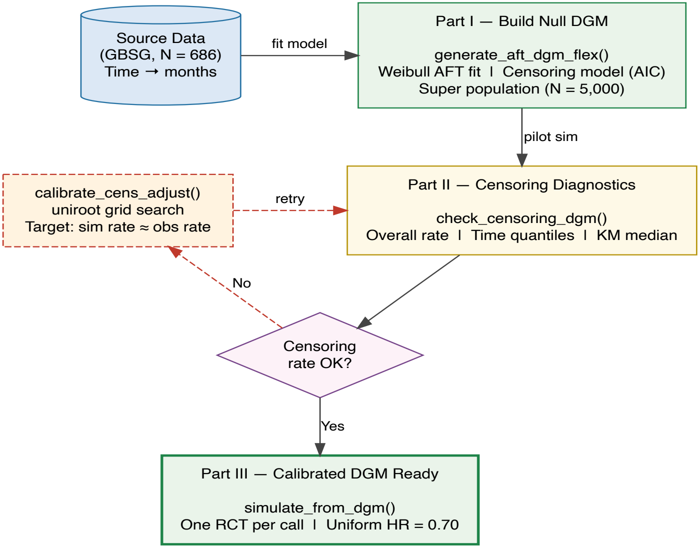
Section 2

Simulation Workflow

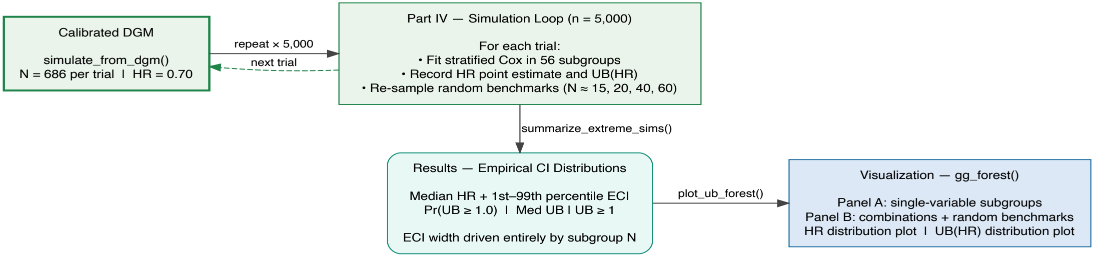
Pipeline Overview

Step	Function	Purpose
1	—	Convert source data to consistent time scale
2	`generate_aft_dgm_flex()`	Fit Weibull AFT; build super population
3	`check_censoring_dgm()`	Verify censoring matches source data
4	`calibrate_cens_adjust()`	Find `cens_adjust` via `uniroot`
5	`simulate_from_dgm()`	Draw synthetic trials from calibrated DGM
6	`coxph()` loop	Fit Cox in 56 subgroups; record HR and UB(HR)
7	`summarize_extreme_sims()`	Compute all summary statistics
8	`plot_ub_forest()`	Visualise ECI distributions

Workflow: DGM Setup & Calibration



Workflow: Simulation Loop & Results



Subgroup Definitions (1/3): ITT, Clinical, Continuous

#	Subgroup	Definition
ITT		
1	All Patients	All subjects (flag_itt == 1)
Clinical		
2	Post-menopausal	meno == 1
3	Pre-menopausal	meno == 0
4	Grade 3	grade == 3
5	Grade 1/2	grade != 3
Continuous		
6	Age (young)	age median
7	Age (older)	age $>$ median
8	Age 50	age 50
9	Age $>$ 50	age $>$ 50
10	Tumour size (small)	size median
11	Tumour size (large)	size $>$ median
12	Node-negative	nodes == 0
13	Nodes 1–3	0 $<$ nodes 3
14	High nodes (>Q3)	nodes > Q3

Subgroup Definitions (2/3): Two-Way Interactions

#	Subgroup	Definition
Meno × Grade		
19	Pre-meno / Grade 3	meno==0 & grade==3
20	Post-meno / Grade 3	meno==1 & grade==3
21	Pre-meno / Grade 1–2	meno==0 & grade!=3
22	Post-meno / Grade 1–2	meno==1 & grade!=3
Meno × Age		
23	Pre-meno / Age 50	meno==0 & age 50
24	Pre-meno / Age > 50	meno==0 & age>50
25	Post-meno / Age (young)	meno==1 & age med
26	Post-meno / Age (older)	meno==1 & age>med
Meno × ER		
27	Pre-meno / ER-low	meno==0 & er 20
28	Pre-meno / ER-high	meno==0 & er>20
29	Post-meno / ER-low	meno==1 & er 20
30	Post-meno / ER-high	meno==1 & er>20
Grade × Nodes		
31	Grade 3 / Node-neg	grade==3 & nodes==0
32	Grade 3 / Node-pos	grade==3 & nodes>0
33	Grade 1–2 / Node-neg	grade!=3 & nodes==0

Subgroup Definitions (3/3): Random Benchmarks

#	Subgroup	Approx N	Construction
53	random60	~60	Draw 60 random indices from trial
54	random40	~40	First 40 of the random60 indices
55	random20	~20	First 20 of the random60 indices
56	random15	~15	First 15 of the random60 indices

Re-sampled every simulation. Nested: random15 random20 random40 random60. No clinical meaning — true HR $=0.70$ by construction.

Section 3

Part I — Building the Null DGM

Fitting `generate_aft_dgm_flex()`

- Fits a **Weibull AFT model** to source data (GBSG)
- Generates a large *super population* ($N_{\text{super}} = 5000$) from which all future trials are drawn
- `model = "null"` — treatment effect identical for every patient
- Censoring model estimated by AIC selection (Weibull / log-normal)

R code

```r
dgm_null <- generate_aft_dgm_flex(
  data = gbsg, model = "null",
  subgroup_vars = NULL,    n_super = 5000)
```

Section 4

Part II — Censoring Diagnostics & Calibration

Censoring Consistency

- Credible results require censoring in synthetic trials to match source data
- `check_censoring_dgm()` compares three quantities:
  1. **Overall censoring rate** vs `rate_tol` (default 10 pp)
  2. **Censoring-time quantiles** among censored subjects
  3. **KM median censoring time** vs `median_tol` (default 25%)

Calibrating `cens_adjust`

`calibrate_cens_adjust()` uses `uniroot` to find the log-scale shift that equates a target statistic between reference and simulated data.

Objective function

At each candidate `cens_adjust`: (1) generate synthetic subjects, (2) extract target metric, (3) return sim_metric − ref_metric.

Target	Equates
`"rate"`	Overall censoring rate
`"km_median"`	KM-based median censoring time

Monotonicity: larger `cens_adjust` ⇒ longer censoring times.

Section 5

Part III — Generating Synthetic Trials

`simulate_from_dgm()`

With the null DGM calibrated, each simulated trial is a single function call:

1. Samples from super population with replacement
2. Assigns treatment ($1{:}1$ randomisation)
3. Generates latent survival times from the AFT model
4. Applies calibrated censoring model
5. Truncates at `analysis_time`

Under the null DGM: `flag_harm` $\equiv 0$ for every subject.

Section 6

Part IV — Simulation Results

Subgroups Under Study (56 total)

Group	Examples
ITT	All Patients ($N \approx 686$)
Clinical	Post-/Pre-menopausal, Grade
Continuous	Age, Size, Nodes, PGR, ER
Interaction	Meno $\times$ Grade, ER, Age; Grade $\times$ Nodes, ER, PGR

Group	Examples
3-way	Pre-meno/Young/ER-low, ...
Size $\times$ Nodes	Large/Node-pos, ...
Random	random60, 40, 20, 15

Key benchmark

`random*` subgroups have **no clinical meaning**. Re-sampled each simulation.

Summary Statistics

HR point estimate distribution

- Median $+$ 1st–99th percentile (ECI)
- Under truth, median ≈ 0.70 for every subgroup
- ECI width driven by subgroup N

Upper confidence bound UB(HR)

- Upper end of the Cox 95% CI
- $\Pr(\text{UB} \geq 2)$, $\Pr(\text{UB} \geq 3)$: tail-risk rates a reviewer would flag
- mUB: median upper bound — the typical worst case per subgroup

Simulation: 1000 trials completed

- **1000** simulated trials from the null DGM (true HR $= 0.70$)
- **56** subgroups evaluated per trial
- Subgroup sizes range from $N \approx 15$ (random15) to $N \approx 686$ (ITT)

Re-use with new results

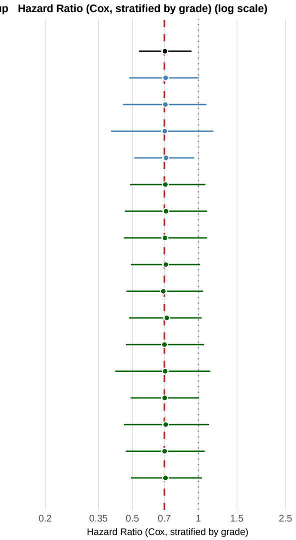
```
res <- readRDS("new_file.rds")
S   <- summary(res, hr_true = 0.70)
```

All downstream tables and plots update automatically.

Section 7

Results — Forest Plots

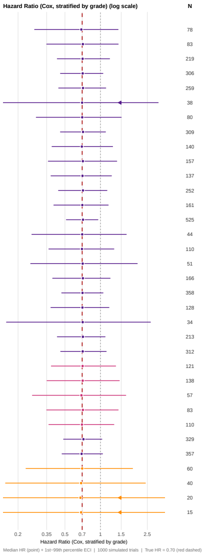
HR Distributions: Single-Variable Subgroups



Subgroup	N	HR<0.5	HR>1.0	mHR
All Patients	686	0.2%	0.2%	0.7
Post−menopausal	389	1.5%	1%	0.71
Pre−menopausal	297	3.6%	2.8%	0.71
Grade 3	161	6.9%	5.1%	0.7
Grade 1/2	525	0.9%	0.5%	0.71
Age (young)	373	1.6%	1.5%	0.71
Age (older)	313	2.5%	2.7%	0.71
Age <= 50	295	2.6%	2.5%	0.7
Age > 50	391	1.1%	1.6%	0.71
Tumour size (small)	357	2.5%	2.1%	0.69
Tumour size (large)	329	1.3%	2.4%	0.72
Nodes 1−3	381	2.8%	2.2%	0.7
High nodes (>Q3)	134	7.8%	5.3%	0.71
PGR low	340	1.8%	1.1%	0.7
PGR high	346	3.3%	3.1%	0.71
ER−low (<20)	277	3.2%	2.3%	0.7
ER−high (>=20)	409	1.2%	1.4%	0.71

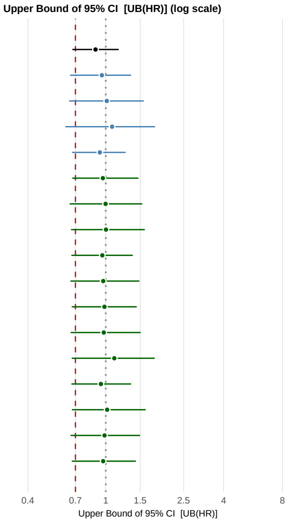
Median HR (point) + 1st−99th percentile ECI | 1000 simulated trials, N = 686 per trial | True HR = 0.70 (red dashed)

HR Distributions: Combinations + Random Benchmarks



Subgroup	N	HR<0.5	HR>1.0	mHR
Pre−meno / Grade 3	78	15.8%	13%	0.69
Post−meno / Grade 3	83	13.9%	13%	0.71
Pre−meno / Grade 1−2	219	4.9%	6.9%	0.71
Post−meno / Grade 1−2	306	3.6%	2.4%	0.71
Pre−meno / Age<=50	259	3.9%	3.4%	0.71
Pre−meno / Age>50	38	29.1%	25.8%	0.7
Post−meno / Age(young)	80	16.4%	14.5%	0.7
Post−meno / Age(older)	309	2.3%	3.1%	0.71
Pre−meno / ER−low	140	8.9%	7.8%	0.7
Pre−meno / ER−high	157	12.2%	10.8%	0.71
Post−meno / ER−low	137	9.7%	7.5%	0.7
Post−meno / ER−high	252	4.3%	4.2%	0.71
Grade 3 / Node−pos	161	6.9%	5.1%	0.7
Grade 1−2 / Node−pos	525	0.9%	0.5%	0.71
Grade 3 / High nodes	44	19.2%	16.3%	0.7
Grade 3 / ER−low	110	11.5%	7.4%	0.7
Grade 3 / ER−high	51	18.3%	20.1%	0.71
Grade 1−2 / ER−low	166	6.5%	7.8%	0.71
Grade 1−2 / ER−high	358	2.4%	2.1%	0.7
Grade 3 / PGR low	128	9.9%	6.9%	0.7
Grade 3 / PGR high	34	28.2%	26.5%	0.71
Grade 1−2 / PGR low	213	5.8%	3.5%	0.71
Grade 1−2 / PGR high	312	4.4%	4.2%	0.71
Pre−meno/Yng/ER−low	121	11.7%	9.4%	0.71
Pre−meno/Yng/ER−high	138	13.4%	12.5%	0.71
Post−meno/G3/ER−low	57	20.4%	16.6%	0.69
Post−meno/G3/Node+	83	13.9%	13%	0.71
G3/Node+/ER−low	110	11.5%	7.4%	0.7
Large/Node−pos	329	1.3%	2.4%	0.72
Small/Node−pos	357	2.5%	2.1%	0.69
random60	60	22.9%	18.7%	0.7
random40	40	27.4%	24.4%	0.69
random20	20	35.1%	32.5%	0.68
random15	15	39.6%	35.9%	0.69

UB(HR) Distributions: Single-Variable Subgroups {.shrink=5}



Subgroup	N	UB>=2	UB>=3	mUB
All Patients	686	0%	0%	0.89
Post−menopausal	389	0%	0%	0.96
Pre−menopausal	297	0%	0%	1.01
Grade 3	161	0.2%	0%	1.08
Grade 1/2	525	0%	0%	0.93
Age (young)	373	0%	0%	0.97
Age (older)	313	0%	0%	1
Age <= 50	295	0%	0%	1
Age > 50	391	0%	0%	0.96
Tumour size (small)	357	0%	0%	0.97
Tumour size (large)	329	0%	0%	0.98
Nodes 1−3	381	0%	0%	0.98
High nodes (>Q3)	134	0.2%	0%	1.11
PGR low	340	0%	0%	0.94
PGR high	346	0%	0%	1.02
ER−low (<20)	277	0.1%	0%	0.99
ER−high (>=20)	409	0%	0%	0.97

Median UB(HR) (point) + 1st−99th percentile ECI | 1000 simulated trials | Dotted line: UB = 1.0; Dashed line: true HR = 0.70

Figure 5: UB(HR) — single-variable subgroups. Dotted: UB=1.0; Dashed: true HR=0.70.

UB(HR) Distributions: Combinations + Random Benchmarks {.shrink=5}

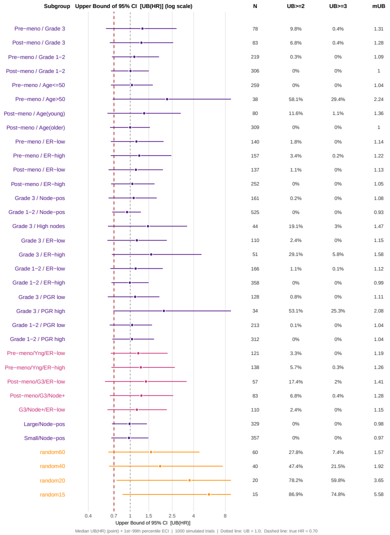


Figure 6: UB(HR) — combination and random-benchmark subgroups.

Section 8

Key Findings

HR Point Estimates

Large subgroups ($N > 200$)

Median HR tracks truth (≈ 0.70) closely; ECI is narrow.

Small subgroups ($N \lesssim 40$)

- ECI for `random15`/`random20` spans ~ 0.25 to > 1.0
- Clinical subgroups of similar size show **indistinguishable** ECI widths
- Clinical rationale provides **no statistical protection**

Clinical subgroups of a given N are statistically indistinguishable from meaningless random groupings of the same size.

Upper Bounds UB(HR)

The upper-bound distribution

- `random40` ($N \approx 40$): 1% of trials produce UB(HR) $>$ 3.0–4.0
- `random15` ($N \approx 15$): values above 5.0 not rare
- A reviewer seeing UB(HR) $=$ 3.5 in 15 patients cannot distinguish chance from a genuine safety signal

Illustrative 99th pctl of UB(HR)

Subgroup	P99 UB
ITT ($N \approx 686$)	< 1.1
$N \approx 200$	~ 1.5
$N \approx 60$	~ 2.5
random40	~ 3.5
random20	~ 5.0
random15	> 5.0

Cox Model Works Correctly

What the simulation confirms

1. Cox model is **unbiased** — median HR ≈ 0.70 for every subgroup
2. Confidence intervals have correct **coverage**
3. Variability driven **entirely by sample size**

The interpretive hazard

- $\mathrm{UB(HR)} = 3.5$ in a 15-patient subgroup is *expected* under uniform benefit
- Without simulation-based calibration, no way to quantify how "extreme" a result is
- The `random*` benchmarks provide exactly this calibration

Section 9

Conclusions

Conclusions

1. **Small subgroups produce extreme results by chance.** Under uniform HR $= 0.70$, subgroups with $N \lesssim 40$ regularly produce UB(HR) > 3.0.
2. **Clinical rationale does not protect against noise.** Clinical subgroups have the *same* HR/UB distribution as random benchmarks of that size.
3. **The Cox model is working correctly.** Wide CIs in small subgroups are accurate — they reflect genuine uncertainty.
4. **Simulation-based calibration is essential.** Without a null-DGM benchmark, reviewers cannot quantify "extreme" for a given subgroup size.
5. **The `forestsearch` package** provides a complete, reproducible workflow.

Thank You

`forestsearch` R package

https://larry-leon.github.io/forestsearch/

```
pak::pak("larry-leon/forestsearch")
```

Section 10

Appendix

New-Results Workflow

```r
library(forestsearch)

#   Option A: Load existing results (payload format)
res <- readRDS("results/extreme_sims_5000.rds")

#   Option B: After a fresh run
# run_subgroup_sims() returns the payload object directly:
saveRDS(sims_uniform, "results/extreme_sims_new.rds")

#   Summarize (works with either option)
S <- summary(res, hr_true = 0.70)

#   All plots driven by S
plot(S, metric = "hr", panel = "single")
plot(S, metric = "hr", panel = "combo")
plot(S, metric = "ub", panel = "single")   # <-- ub-forest-single
plot(S, metric = "ub", panel = "combo")    # <-- ub-forest-combo
```

summary() Fields

Field	Description
hr_q, ub_q	Quantile matrices [n_sg $\times$ 3]: 1st, median, 99th
mean_n, n_per_trial	Mean subgroup N; per-trial N (footnotes)
pr_hr_lt050, pr_hr_gt1	HR tail probabilities per subgroup
pr_ub_ge2, pr_ub_ge3	UB tail-risk rates per subgroup
results_tbl	Formatted data.frame for printing
ok, ok_ub	Logical: valid subgroups
idx_hr_single, idx_ub_combo, ...	Index vectors for forest panels
cat, cat_cols	Category labels and colour map
highrisk	High-risk panel: threshold, membership, ordering

Comparing Two Simulation Runs

```r
res_1k <- readRDS("results/extreme_sims_1000.rds")   # payload objects
res_5k <- readRDS("results/extreme_sims_5000.rds")

cmp <- compare_subgroup_sims(res_1k, res_5k,
                             suffixes = c("_1k", "_5k"))
head(cmp[, c("subgroup", "N_1k", "N_5k", "ub2_1k", "ub2_5k",
             "mHR_1k", "mHR_5k")])
# Same function the fixed-vs-random design-comparison memo uses
# (expect_designs = c("resample", "fixed") guards that pairing).
```

Useful for verifying convergence, comparing DGM variants, or assessing sensitivity to censoring calibration.

Helper Utilities

```r
#   Optimal figure height
fh <- forest_height(S, panel = "combo",
                    row_height = 0.45, overhead = 1.5)

#   Save / load round-trip (payload format)
saveRDS(sims_uniform, "results/my_run.rds")
res <- readRDS("results/my_run.rds")   # class "subgroup_sims" intact
print(res)  # design, n_sims, subgroups, fit, provenance

#   Print method
print(S)   # header (design | n_sims | true HR), then results_tbl
```